

# **Dwelling on Dwell Times: Evaluating the Relative Importance of Features Associated with Passenger Train Dwell Times on the Belgian Rail Network**

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## **Abstract**

A Monte Carlo simulation previously developed to quantify the impact of signal noise on performance metrics for the Belgian passenger rail network consistently underestimated train dwell times and suffered from simulation stability issues relating to data subsetting. To better understand the factors associated with dwell times and identify variables of interest for data subsetting, potentially relevant predictors of interest based on a review of prior literature and categorical equivalents likely useful in data subsetting for simulations were calculated from granular event data published by Belgian public rail network operator Infrabel. Four regression estimators—XGBoost Regressor, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor—were fitted to two sets of predictors: one consisting of an unmodified collection of continuous and categorical predictor variables and the other consisting of a mixture of the continuous predictor variables reduced to binary categorical variables and the original categorical predictor variables. None of these regression models performed satisfactorily at predicting train dwell times; however, the Gini feature importances output from each model identified (1) a single continuous predictor that remained important after being converted to a binary categorical predictor, (2) several categorical temporal characteristics that were consistently irrelevant to both sets of regression models, and (3) a series of relevant continuous predictors that, upon being converted to binary categorical predictors, became largely irrelevant. These results suggest that there is promise in this approach to variable selection for simulation development; however, further work needs to be done in exploratory analyses and understanding how information loss manifests.

## **Introduction**

Over the course of August and September of last year, over two-hundred-thousand unique passenger train runs were completed on the Belgian passenger rail network. Automated systems tracked these trains as they completed their routes, recording each time they arrived at and departed from a measurement point (either a rail junction or train station). Data on these arrival and departure events are published monthly by Belgian public rail network infrastructure operator Infrabel; because these events are listed sequentially for each unique passenger train run, travel times between stations and dwell times at stations can be extracted. However, calculating these values reveals a major problem: excessively low and high train dwell and travel times indicative of measurement error that could adversely impact the calculation of performance indicators for train drivers, train operators, and (more broadly) network operations.

To understand the impact of signal noise on two possible performance metrics—average arrival delay and arrival delay at the end of the route—a Monte Carlo simulation was developed and evaluated that consistently (a) underestimated dwell times at stations, (b) overestimated travel times between stations, and (c) suffered from stability issues relating to data subsetting.

These issues resulted from (a) a lack of prior research on train dwell and travel times and (b) poor data cleaning. While the simulations that successfully ran yielded non-statistically significant differences from the equivalent true performance metrics in a paired samples t-test at significance level  $\alpha = 0.05$ , this result was biased towards the simulations that ran successfully and was likely underpowered due to the small sample size ( $n = 44$ ).

This project aims to refine this simulation-based approach by training a collection of regression models using an assortment of relevant estimators—XGBoost, Decision Tree, Random Forest, and Gradient Boosting— and extracting the Gini feature importances from each trained model to perform an analysis of and better understand the underlying factors influencing passenger train dwell times on the Belgian passenger rail network. In so doing, this project further seeks to evaluate the potential upsides and shortcomings of this methodology, more broadly, as a framework for variable selection and operationalization for Monte Carlo simulations.

## Methodology

### 1. Data Extraction

This project necessitated the use of three datasets: (1) granular event data for August 2025 with approximately 1.9 million records and 21 variables, (2) granular event data for September 2025 with approximately 2 million records and 22 variables, and (3) data consisting of the distances between all adjacent pairs of train stations in Belgium with 1,385 records and 7 variables; all three datasets were sourced from Infrabel’s Open Data portal. The first and second datasets are necessary to calculate dwell times as well as the operationalized forms of all relevant predictors (see table 1 below). Dataset 3 was required to (a) subset for valid train stations (as opposed to non-station measurement points, such as junctions) and (b) calculate the distances and average velocities of trains between stations to help identify and remove outliers.

**Table 1.** A list of predictors calculated from the source datasets with the associated variable name in the dataset (the “label”) and data type (discrete, continuous, ordinal, or nominal). Predictors 2A, 2B, 2C, 2D, and 2I are binary operationalized forms of predictors 1A, 1B, 1C, 1D, and 1I, respectively.

| Label     | Predictor                                                                                             | Data Type                |
|-----------|-------------------------------------------------------------------------------------------------------|--------------------------|
| <b>1A</b> | The difference between the planned and actual dwell time at the preceding station.                    | Continuous (Seconds)     |
| <b>1B</b> | The difference between the planned and actual dwell time for the preceding train at the same station. | Continuous (Seconds)     |
| <b>1C</b> | The difference between the planned and actual travel time on the preceding track segment.             | Continuous (Seconds)     |
| <b>1D</b> | The time since the previous valid train arrived at the same station.                                  | Continuous (Seconds)     |
| <b>1E</b> | Arrival time binned on 2-hour intervals.                                                              | Ordinal (Bin start hour) |
| <b>1F</b> | ...on 3-hour intervals.                                                                               | Ordinal (Bin start hour) |

|           |                                                                                                                          |                               |
|-----------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>1G</b> | ...on 4-hour intervals                                                                                                   | Ordinal (Bin start hour)      |
| <b>1H</b> | Train operator                                                                                                           | Nominal (Company)             |
| <b>1I</b> | Day of the week on which the train initially departed                                                                    | Ordinal (Numeric day of week) |
| <b>2A</b> | Whether the train dwelled longer than planned at the preceding station.                                                  | Ordinal (Binary)              |
| <b>2B</b> | Whether the preceding valid train at the same station dwelled for longer than expected.                                  | Ordinal (Binary)              |
| <b>2C</b> | Whether the train took longer than expected to traverse the preceding track segment.                                     | Ordinal (Binary)              |
| <b>2D</b> | Whether the time since the previous valid train stopped at the same station is in the upper 50 <sup>th</sup> percentile. | Ordinal (Binary)              |
| <b>2I</b> | Whether the train initially departed on a weekend.                                                                       | Ordinal (Binary)              |

## 2. Data Preparation

### A. Feature Engineering

Datasets 1 and 2 (granular event data for August and September 2025) were combined row-wise after the “OP1\_COD” variable was dropped from the latter dataset (the dropped variable was irrelevant to this analysis). Planned and actual arrival and departure Pandas datetime objects were created for all events in the combined dataset. From these Pandas datetime objects, planned and actual dwell time at train stations were calculated as the difference between planned departure and arrival time and actual departure and arrival time, respectively, in seconds; the latter represents the actual dwell time and will be referenced as response variable “y.” Additionally, the differences between planned and actual departure and arrival times, in seconds, were calculated. Missing values for the difference between actual and planned arrival time for the initial departure event on each train’s route (where there is no arrival timestamp) were imputed with a value of 0 seconds; this imputation supported the subsequent calculation of the difference between actual and planned dwell time as the change in the difference between actual and planned departure times and the difference between actual and planned arrival times.

At this point, dataset 3, documenting the distances between adjacent stations, was processed: (1) station pairs (and their distances) were copied, flipped, and appended to the end of the table; (2) duplicate pairs of stations were dropped, which added one pair of stations that had not been accounted for in the original dataset; (3) a list of unique stations was created; and (4) the main combined dataset was subset for rows with measurement point IDs in the list of unique stations, indicating each was itself a station, rather than a junction. Afterwards, a *for* loop iterated through the dataset appending a unique run ID (a 1-indexed integer iterated sequentially by 1) to each event for each train run present in the dataset; this was possible because, for each train run in the dataset, relevant events are listed sequentially.

Using these run IDs, the actual and planned travel times between stops, in seconds, were calculated for each individual run of each train. For each station at which a train wasn’t

scheduled to stop, a unique stop ID associated with the next planned stop on the train's route was appended; this 0-indexed value iterated sequentially by 1 after each planned stop. The unique measurement point ID for the preceding train station was added for each non-origin station. After adding these features to the combined dataset, (1) all missing actual travel times on the preceding station-to-station segment were dropped, removing all origin departure events; (2) dataset 3 was left-joined on the origin and destination measurement point IDs; (3) the total travel time and distances between stops were summed for each arrival stop, and stations with planned and actual dwell times less-than-or-equal-to 0 were dropped; and (4) average velocities for each stop-to-stop segment were calculated in kilometers per hour (km/h). The day of the week as an ordinal, categorical value (1-7, beginning with Monday), based on the origin departure time for each train run—predictor *II* (binary operationalized as predictor *2I*; see table 1)—was calculated and appended for each event

The difference between planned and expected dwell times at the preceding station and the difference between planned and expected travel times on the preceding stop-to-stop segment—predictors *IA* and *IC* (see table 1), respectively—were calculated by applying a *for* loop to each unique run of each train; the binary operationalized variations of these variables—predictors *2A* and *2C*, respectively—were calculated from these values. Arrival time bins in two-, three-, and four-hour bin sizes—predictors *IE*, *IF*, and *IG* (see table 1), respectively—were appended to the dataset at this stage as well. To remove potential outlier station-to-station travel times, (1) the average stop-to-stop velocities were divided by the total travel time between stops to penalize extreme high and low average velocities achieved over short periods of time (e.g., it is highly unrealistic for a train to achieve an average velocity of 250 km/h over a 40-second travel time between stations); and (2) the upper and lower 0.1% of travel-time-penalized velocities were removed.

Subsequently, the data were grouped by arrival station ID and sorted in ascending order by actual arrival time. Iterating over data for each station, (1) the difference between the arrival time for each train and that of the preceding train at the same station and (2) the difference between actual and planned dwell time for the preceding train at the same station—predictors *IB* and *ID* (see table 1), respectively—were calculated; from these values, the binary operationalized forms of each variable—predictors *2B* and *2D*, respectively—were calculated. Missing values for predictors *IB* and *ID*, which logically only occur for the first valid event for each station, were removed.

After all relevant predictors were calculated or appended to the dataset, it was split in two (see variables selected in table 2 below; variable names introduced in table 1): the first dataset, consisting of a mixture of continuous and categorical variables, retains all information in the original dataset and represents the best possible conditions (without subsetting for specific stations, routes, or types of train lines) for a regression model; the second dataset represents a more realistic set of subsetting variables for the Monte Carlo simulation approach previously attempted to (a) identify potentially relevant variables for simulation and (b) better understand the information loss from the continuous variables to their binary encoded counterparts.

**Table 2.** *Predictor variables by dataset. The ensemble of regression estimators will be applied to each dataset after continuous values have been normalized and categorical variables with more*

than two levels have been one-hot encoded. Variables shared across both datasets are in merged cells.

| Processed Dataset 1<br>(Continuous/Categorical) | Processed Dataset 2<br>(Categorical) |
|-------------------------------------------------|--------------------------------------|
| <i>y</i> (Target)                               |                                      |
| <i>1A</i>                                       | <i>2A</i>                            |
| <i>1B</i>                                       | <i>2B</i>                            |
| <i>1C</i>                                       | <i>2C</i>                            |
| <i>1D</i>                                       | <i>2D</i>                            |
| <i>1E</i>                                       |                                      |
| <i>1F</i>                                       |                                      |
| <i>1G</i>                                       |                                      |
| <i>1H</i>                                       |                                      |
| <i>1I</i>                                       | <i>2I</i>                            |

### B. Normalization and Encoding

Continuous predictors were normalized with standard scaler, with the means and standard deviations for each predictor, to compute the z-scores of each value, set to those of the training set for dataset 1 and subsequently applied to both the training and test sets for dataset 1. One-hot encoding (otherwise known as dummy coding) was applied to the categorical predictors with more than two levels—predictors *1E*, *1F*, *1G*, *1H*, and *1I*. No normalization or encoding was required for predictors *2A*, *2B*, *2C*, *2D*, and *2I* due to their already being binary (i.e., having either true, 1, or false, 0, values).

## 3. Data Exploration

Exploratory data analysis (EDA) was performed in two phases: univariate and bivariate analysis. For continuous variables, a tabular representation of summary statistics produced using Pandas (specifically `pd.DataFrame.describe()`) and histograms were employed for univariate analysis; for bivariate analysis, a correlation heatmap was produced using Seaborn (. For categorical variables, a univariate analysis was performed using bar charts to assess the distribution of

### A. Univariate Analysis

**Table 3.** The five number summary, mean, and standard deviation reported for all five continuous variables in dataset 1 (four continuous predictors and the response).

|              | <i>y</i> | <i>1A</i> | <i>1B</i> | <i>1C</i> | <i>1D</i> |
|--------------|----------|-----------|-----------|-----------|-----------|
| <b>Count</b> | 1619559  | 1619559   | 1619559   | 1619559   | 1619559   |
| <b>Mean</b>  | 115.085  | 21.545    | 15.583    | -17.904   | 1694.708  |
| <b>SD</b>    | 121.372  | 88.159    | 73.076    | 65.244    | 8687.015  |
| <b>Min</b>   | 1        | -1797     | -1797     | -4602     | 0         |
| <b>25%</b>   | 57       | -9        | -11       | -46       | 200       |

|               |      |      |      |      |         |
|---------------|------|------|------|------|---------|
| <b>Median</b> | 81   | 11   | 7    | -19  | 646     |
| <b>75%</b>    | 127  | 38   | 35   | 7    | 1606    |
| <b>Max</b>    | 9791 | 9047 | 9551 | 2194 | 1844113 |

Per table 3 above, the response variable  $y$  appears to be heavily right-skewed, with the mean dwell time (approximately 115.09 seconds) far exceeding the median (81 seconds) and being just shy of the third quartile (127 seconds); in addition, the difference between the maximum dwell time (9,971 seconds) and third quartile (127 seconds) far exceeding the difference between the first quartile (57 seconds) and the minimum (1 second). A similar pattern appears in predictor variable  $ID$ , where the mean time since the last valid train arrived at the same station (approx. 1,694.71 seconds) far exceeds the median (646 seconds) and even exceeds the third quartile (1,606 seconds); similarly, the difference between the maximum value (approx. 1.8 million seconds) and the third quartile (1,606 seconds) far exceeds the difference between the first quartile (200 seconds) and the minimum value (0 seconds).

Different trends hold for continuous predictors  $IA$ ,  $IB$ , and  $IC$ , where the means (approx. 21.55, 15.58, and -17.9 seconds, respectively) are approximately equal to the medians (11, 7, and -19 seconds, respectively) relative to the range of all three variables. While predictors  $IA$  and  $IB$  appear to exhibit a slight right skew, with the distance between their third quartiles (38 and 35 seconds, respectively) and respective medians (11 and 7 seconds, respectively) exceeding the distance between their medians and first quartiles (-9 and -11 seconds), predictor  $IC$  appears to exhibit a slight left skew with the difference between its median (-19 seconds) and first quartile (-46 seconds) exceeding the distance between the third quartile (7 seconds) and median (-19 seconds). Due to the apparent heaviness of their tails and slight skewness, all three of these variables appear to be approximately symmetric but leptokurtic (i.e., with heavier tails than might be expected from a Normal distribution).

*Table 4. The count, number of unique levels, most frequent level, and the count frequency of the most frequent level for all five categorical predictors in dataset 1 (1E, 1F, 1G, 1H, and 1I).*

|               | <b>1E</b> | <b>1F</b> | <b>1G</b> | <b>1H</b> | <b>1I</b> |
|---------------|-----------|-----------|-----------|-----------|-----------|
| <b>Count</b>  | 1619559   | 1619559   | 1619559   | 1619559   | 1619559   |
| <b>Unique</b> | 12        | 8         | 6         | 3         | 7         |
| <b>Top</b>    | 16        | 15        | 16        | SNCB/NMBS | 1         |
| <b>Freq</b>   | 206030    | 297427    | 397535    | 1615373   | 280445    |

*Table 5.*

|               | <b>2A</b> | <b>2B</b> | <b>2C</b> | <b>2D</b> | <b>2I</b> |
|---------------|-----------|-----------|-----------|-----------|-----------|
| <b>Count</b>  | 1619559   | 1619559   | 1619559   | 1619559   | 1619559   |
| <b>Unique</b> | 2         | 2         | 2         | 2         | 2         |
| <b>Top</b>    | TRUE      | TRUE      | FALSE     | FALSE     | FALSE     |
| <b>Freq</b>   | 1042599   | 990894    | 1121620   | 810510    | 1313536   |

## *B. Bivariate Analysis*

### 4. Predictive Modeling

Since the target variable, dwell time (in seconds), is continuous, this is a regression problem. Tree-based regression methods—such as XGBoost Regression and Gradient Boosting Regression—were recommended in the context of variable selection for dwell time estimation by Tiong et al. (2023) because they (a) provide Gini feature importances from model training unlike an RNN, which may provide superior inference per Wen et al. (2019) but lack a similar, easy-to-interpret metric for the underlying importance of predictors, and (b) are theoretically able to handle the complex nested relationships inherent to large, real-world operations datasets. The ensemble of tree-based estimators applied in this analysis—XGBoost Regression, Gradient Boosting Regression, Random Forest Regression, and Decision Tree Regression—was applied to both processed datasets (see table 2) for a total of eight models from which to assess the relative feature importances for predictors of dwell time.

### 5. Evaluation Metrics

Four evaluation metrics— $R^2_{\text{adj}}$ , Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE)—were selected to assess the inferential performance of each regression model trained.  $R^2_{\text{adj}}$ —the percentage of variance in the response accounted for by a regression model—is the most common metric by which all regression models are measured; in this context, it was used as the scoring metric for grid search cross validation for all models trained. MAE and RMSE are similar metrics conceptually; however, MAE, calculated as the average absolute value of error, is more resistant to outliers than RMSE which, being calculated as the square root of the population variance of error for each regression model (MSE), is much more sensitive to outliers. Both provide an intuitive measure of model fit by representing average error in units of the response (seconds, in this case), which should (and will) be discussed in the broader context of the distribution of the response/target variable. Finally, MAPE—the average of the absolute value of the percentage by which predictions deviate from actual values—adds further context to the MAE value and is commonly applied to that end in assessing regression models.

## **Experiments**

### 1. Experimental Datasets

Three datasets were retrieved from the Infrabel Open Data portal: (1) a dataset with 1,861,871 records across 21 variables containing granular event data from August 2025, (2) a dataset with 2,015,606 records across 22 variables containing granular event data from September 2025, and (3) a dataset with 1,385 records across 7 variables containing the distances between pairings of adjacent stations; the first and second datasets were useful in calculating all spatiotemporal characteristics that could be calculated from this data and that could theoretically be relevant in predicting train dwell times, while the third dataset was useful for calculating extreme high and low velocities and removing outliers.

This data was subsequently transformed and cleaned into 1,619,559 records across 15 variables (14 predictors and 1 response variable). Two experimental datasets were derived from the transformed and cleaned data (variables enumerated in table 2): processed dataset 1, which contained a mix of continuous and categorical variables (full information); and processed dataset 2, which only contained categorical variables (reduced information). Processed dataset 1 and its derived models were used to assess the relevance of all of the predictor variables with full information; processed dataset 2 and its derived models were employed to assess the impacts of information loss from the collapsing of continuous predictors *IA*, *IB*, *IC*, and *ID* and continuous predictor *II* (day of the week) into binary categorical predictors. Both datasets had 1,619,559 records after cleaning across 10 variables (9 predictors and the response variable).

The data was split using a highly unconventional 10-90 train-test split; this was deemed necessary due to (1) the limited time available to complete model training, (2) extremely limited computational resources available given the scale and complexity of the dataset, and (3) a desire to increase the number of cross-validation folds make up for the lack of training data by validating hyperparameters against a wider diversity of data subsets. Using the column transformer from scikit-learn (`sklearn.compose.ColumnTransformer()`) to intuitively define a transformation pipeline, processed dataset 1 predictor variables *IA*, *IB*, *IC*, and *ID* were normalized and *IE*, *IF*, *IG*, *IH*, and *II* were one-hot encoded. Processed dataset 2 variables *IE*, *IF*, *IG*, and *IH* were one-hot encoded; the rest of the predictor variables did not require additional transformation for model parameter tuning, training, and evaluation.

These transformations yielded eight matrices: (1) the prediction training split for processed dataset 1 with 161,955 records across 40 columns; (2) the prediction test split for processed dataset 1 with 1,457,604 records across 40 columns; (3) the target training split for processed dataset 1 with 161,955 records across 1 column; (4) the target test split for processed dataset 1 with 1,457,604 records across 1 column; (5) the prediction training split for processed dataset 2 with 161,955 records across 34 columns; (6) the prediction test split for processed dataset 2 with 1,457,604 records across 34 columns; (7) the target training split for processed dataset 2 with 161,955 records across 1 column; and (8) the target test split for processed dataset 2 with 1,457,604 records across 1 column. Both train-test splits were performed with random state 102, and variable normalization for continuous predictors was based on the mean and standard deviation of each variable in the prediction training split for processed dataset 1.

## 2. Parameter Tuning

Each training set of 161,955 records was split into 20 folds for grid search cross-validation, which yielded cross-validation folds with approximately 8,097 records per fold; an *a priori* power analysis with G\*Power (F-tests; “Linear multiple regression: Fixed model,  $R^2$  deviation from zero”; an expected effect size of 0.2; a target power of 0.95 corresponding to an  $\alpha$  of 0.05; and 40 predictors, corresponding to the total width of the training matrices) determined the minimum sample size for each individual cross-validation fold to be 228 records, indicating each cross-validation fold, despite its limited size, is likely representative of the population of dwell times on the Belgian rail network. Optimal parameters were chosen based solely on the results of this grid search; this is *not* optimal and will be revised in any follow-on analysis to this project. Table 6 below summarizes the parameter grid for each estimator (the parameter grids were the same for both sets of models to simplify parameter tuning).



**Table 6.** The parameter grid and optimal parameters returned from 20-fold grid search cross-validation for each regressor for each model trained. The continuous/categorical parameters were cross-validated on processed dataset 1; the categorical parameters were cross-validated on processed dataset 2.

| Estimator                          | Parameter Grid                                                                                                                   | Optimal Parameters                                                                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>XGBoost Regressor</b>           | <ul style="list-style-type: none"> <li>"n_estimators": [10, 20, 40, 80, 160],</li> <li>"max_depth": [2, 4, 8, 16, 32]</li> </ul> | Continuous/Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 10</li> <li>"max_depth": 8</li> </ul>   |
|                                    |                                                                                                                                  | Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 20</li> <li>"max_depth": 4</li> </ul>              |
| <b>Decision Tree Regressor</b>     | <ul style="list-style-type: none"> <li>"max_depth": [2, 4, 8, 16, 32]</li> </ul>                                                 | Continuous/Categorical: <ul style="list-style-type: none"> <li>"max_depth": 8</li> </ul>                               |
|                                    |                                                                                                                                  | Categorical: <ul style="list-style-type: none"> <li>"max_depth": 8</li> </ul>                                          |
| <b>Random Forest Regressor</b>     | <ul style="list-style-type: none"> <li>"n_estimators": [10, 20, 40, 80, 160],</li> <li>"max_depth": [2, 4, 8, 16, 32]</li> </ul> | Continuous/Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 160</li> <li>"max_depth": 16</li> </ul> |
|                                    |                                                                                                                                  | Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 40</li> <li>"max_depth": 8</li> </ul>              |
| <b>Gradient Boosting Regressor</b> | <ul style="list-style-type: none"> <li>"n_estimators": [10, 20, 40, 80, 160],</li> <li>"max_depth": [2, 4, 8, 16, 32]</li> </ul> | Continuous/Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 40</li> <li>"max_depth": 8</li> </ul>   |
|                                    |                                                                                                                                  | Categorical: <ul style="list-style-type: none"> <li>"n_estimators": 40</li> <li>"max_depth": 4</li> </ul>              |

### 3. Results

#### A. Evaluation Metrics

**Table 7.** Relevant evaluation metrics for the four regression models fitted to a combination of continuous and categorical predictors (processed dataset 1) with the highest value for each metric **highlighted** and **bolded**.

| Estimator                     | $R^2_{adj}$ | MAE    | MAPE    | RMSE    |
|-------------------------------|-------------|--------|---------|---------|
| XGBoost Regressor (XGBR)      | 16.215%     | 55.767 | 65.983% | 111.39  |
| Decision Tree Regressor (DTR) | 10.732%     | 57.242 | 66.983% | 114.977 |
| Random Forest Regressor (RFR) | 15.787%     | 56.253 | 65.490% | 111.675 |

|                                   |                |               |                |                |
|-----------------------------------|----------------|---------------|----------------|----------------|
| Gradient Boosting Regressor (GBR) | <b>16.244%</b> | <b>55.629</b> | <b>64.684%</b> | <b>111.371</b> |
|-----------------------------------|----------------|---------------|----------------|----------------|

**Table 8.** Relevant evaluation metrics for the four regression models fitted to categorical predictors derived from the potentially relevant continuous predictors (processed dataset 2) with the highest value for each metric **highlighted** and **bolded**.

| Estimator                         | $R^2_{adj}$   | MAE           | MAPE           | RMSE           |
|-----------------------------------|---------------|---------------|----------------|----------------|
| XGBoost Regressor (XGBR)          | 6.860%        | <b>61.907</b> | <b>74.736%</b> | 117.444        |
| Decision Tree Regressor (DTR)     | 5.995%        | 62.058        | 74.744%        | 117.989        |
| Random Forest Regressor (RFR)     | 6.606%        | 61.997        | 74.810%        | 117.605        |
| Gradient Boosting Regressor (GBR) | <b>6.864%</b> | 61.958        | 75.001%        | <b>117.442</b> |

Overall, per tables 7 and 8, the set of regression models fitted to the mixture of continuous and categorical predictors in processed dataset 1 outperformed the set of regression models trained solely on the categorical predictors in processed dataset 2 on all metrics assessed; however, neither set of models performs especially well at prediction (with the models trained with Gradient Boosting Regressor, the best outcome for both sets of predictors, accounting for approx. 16.244% and 6.864% of variance in the dwell time, respectively). To put the difference between these models into perspective, the mixture of continuous and categorical predictors only reduces mean absolute error (MAE)—the average absolute value of error—by approx. 5-6 seconds compared to the dataset consisting only of categorical predictors against an average dwell time of approx. 115.1 seconds. Additionally, MAPE—mean absolute percentage error—hovers between 64% and 67% for the models fitted to processed dataset 1 indicating that, on average, predictions deviate from actual dwell times by around 64-67% of the value of the latter; while this worsens to 74-75% for the models trained solely on categorical predictors, both values, when coupled with root mean square error (RMSE) values for both models that hover around the mean of the response, strongly suggest these models are similarly weak, despite the headline adjusted R-squared values.

Low adjusted R-squared values were expected in this context due to three key features of this dataset and the feature engineered predictors: (1) the scale of automated data collection means measurement error is not only present but is highly prevalent in this dataset; (2) a one-sized fits all regression model is meaningless when meaningful trends likely only extend to an individual station on an individual line at a certain time of day; (3) several predictor variables in this dataset (the arrival time bins and preceding delays) were potentially collinear, which could have led to increased variance in coefficients, an artificially attenuated relationship between individual predictors and the response, and type II error; and (4) variables *1B* and *2B*, recording the difference between actual and expected dwell time for the preceding valid train at the same station, constitute a violation of independence. The weaknesses inherent in this approach to predictive modeling were tolerated in this analysis because the predictions themselves are not relevant to its purpose; the real topic of interest is the Gini feature importances reported by both ensembles of regression models (see tables 9 and 10 below).

## B. Gini Feature Importances

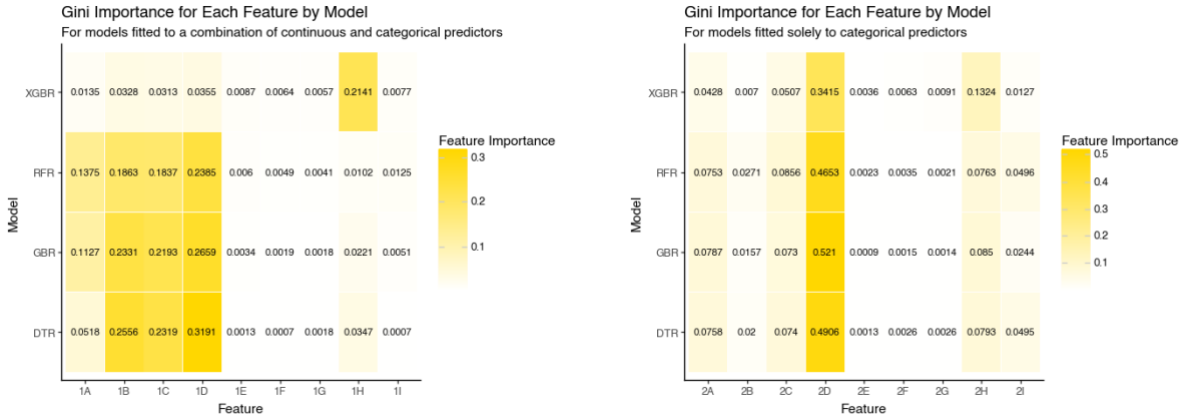
**Table 9.** Gini importance for each predictor by fitted regression model for the models trained on a combination of continuous and categorical predictors, with top 3 values for each model **bolded** and highest value **highlighted**. The asterisk after several of the predictors indicates the feature importance was averaged across all one-hot encoded levels of the variable.

| Model                                                                                                             | XGBR          | DTR           | RFR           | GBR           |
|-------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|
| <b>Predictor (Processed Dataset 1)</b>                                                                            |               |               |               |               |
| (1A) Difference between planned and actual dwell time at preceding station (seconds)                              | 0.0135        | 0.0518        | 0.1375        | 0.1127        |
| (1B) Difference between planned and actual dwell time for the preceding valid train at the same station (seconds) | <b>0.0328</b> | <b>0.2556</b> | <b>0.1863</b> | <b>0.2331</b> |
| (1C) Difference between planned and actual travel time on preceding track segment (seconds)                       | 0.0313        | <b>0.2319</b> | <b>0.1837</b> | <b>0.2193</b> |
| (1D) Time since the previous valid train stopped at the same station (seconds)                                    | <b>0.0355</b> | <b>0.3191</b> | <b>0.2385</b> | <b>0.2659</b> |
| (1E) Arrival time binned on 2-hour intervals (bin start hour)*                                                    | 0.0087        | 0.0013        | 0.006         | 0.0034        |
| (1F) ... on 3-hour intervals*                                                                                     | 0.0064        | 0.0007        | 0.0049        | 0.0019        |
| (1G) ... on 4-hour intervals*                                                                                     | 0.0057        | 0.0018        | 0.0041        | 0.0018        |
| (1H) Train Operator (Company)*                                                                                    | <b>0.2141</b> | 0.0347        | 0.0102        | 0.0221        |
| (1I) Day of the week of the initial departure timestamp (Day)*                                                    | 0.0077        | 0.0007        | 0.0125        | 0.0051        |

**Table 10.** Gini importance for each predictor by fitted regression model for the models trained entirely on categorical predictors, with top 3 values for each model **bolded** and highest value **highlighted**. The asterisk after several of the predictors indicates the feature importance was averaged across all one-hot encoded levels of the variable.

| Model                                                                                                                     | XGBR          | DTR           | RFR           | GBR           |
|---------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|
| <b>Predictor (Processed Dataset 2)</b>                                                                                    |               |               |               |               |
| (2A) Whether the train dwelled longer than planned at preceding station (0 or 1)                                          | 0.0428        | <b>0.0758</b> | 0.0753        | <b>0.0787</b> |
| (2B) Whether the preceding valid train at the same station dwelled longer than expected (0 or 1)                          | 0.007         | 0.02          | 0.0271        | 0.0157        |
| (2C) Whether the train took longer than expected to traverse the preceding track segment (0 or 1)                         | <b>0.0507</b> | 0.074         | <b>0.0856</b> | 0.073         |
| (2D) Whether the time since the previous valid train stopped at the same station is in the upper 50th percentile (0 or 1) | <b>0.3415</b> | <b>0.4906</b> | <b>0.4653</b> | <b>0.521</b>  |
| (1E) Arrival time binned on 2-hour intervals*                                                                             | 0.0036        | 0.0013        | 0.0023        | 0.0009        |

|                                                                |               |               |               |              |
|----------------------------------------------------------------|---------------|---------------|---------------|--------------|
| (1F) ... on 3-hour intervals*                                  | 0.0063        | 0.0026        | 0.0035        | 0.0015       |
| (1G) ... on 4-hour intervals*                                  | 0.0091        | 0.0026        | 0.0021        | 0.0014       |
| (1H) Train Operator (Company)*                                 | <b>0.1324</b> | <b>0.0793</b> | <b>0.0763</b> | <b>0.085</b> |
| (2I) Whether the train initially departs on a weekend (0 or 1) | 0.0127        | 0.0495        | 0.0496        | 0.0244       |



**Figure 1.** Left: An annotated heatmap displaying the Gini importance for each feature by each model fitted to a combination of continuous and categorical predictors. Right: An annotated heatmap displaying the Gini importance for each feature by each model fitted solely to categorical predictors. Variables 2E, 2F, 2G, and 2H are aliases (for ease of understanding in the poster) for predictor variables 1E, 1F, 1G, and 1H, respectively.

Boarding and alighting events were consistently referenced in the literature as the most important factor determining train dwell times. Because Infrabel does not publish granular data on these events, the time since the most recent train arrived at the same station, predictor *1D*, was devised as a proxy for passenger demand at a given time and station and, in turn, as a proxy for boarding and alighting events. That *1D* and its categorical equivalent *2D* are consistently important to both sets of regression models (except for XGBoost regressor fitted to processed dataset 1) indicates both successfully served as (imperfect) proxy variables. Additionally, that the variable remained important despite its being reduced to a binary predictor in processed dataset 2 indicates there was minimal information loss and that a simulation employing variable *2D* as a subsetting variable may be able to account for a substantial amount of variance in train dwell times with a mere two levels.

The same cannot be said for the other continuous predictors in processed dataset 1: *1A*, *1B*, and *1C*. Though *1A* was less relevant as a continuous predictor than the other two, all three were consistently important relative to all but one of the categorical predictors for three of the four regression models trained on processed dataset 1; however, all three dropped off in relative importance for the set of regression models trained on processed dataset 2, likely due to information loss in their conversion from continuous variables to binary categorical variables. Consequently, predictor *1H* (the train operator) became more important relative to the other predictors for all four regression models trained on processed dataset 2, perhaps due to some combination of information being retained across datasets *and* a lack of collinearity with any of the other predictors. This multicollinearity, as previously noted, is the likely culprit for arrival

time bins *IE*, *IF*, and *IG* being relatively “unimportant” across both training datasets and model ensembles. Additionally, both day of week features *II* and *2I* were less relevant to both ensembles of models than expected; however, a more thorough analysis of trends in both of these variables through further univariate and bivariate analyses is required before attempting to identify a root cause for their irrelevance to both sets of regression models.

## Conclusions

Training two sets of regression models on two separate training datasets as part of a variable selection procedure for a Monte Carlo simulation may seem a fool’s errand, but the inclusion of both sets of predictors in this analysis has yielded a far more interesting set of results and possible outcomes than may otherwise have been possible by merely applying the ensemble of regression models to a single set of variables. By including a baseline of variables with full information, it was possible to identify the consistent relevance of the time since the most recent “valid” arrival at the same station (feature *ID*), both as a continuous predictor for regression and as a categorical variable that is potentially relevant for simulation design. Additionally, this analysis identified (a) potential variables of interest that, while relevant as continuous predictors, suffered from information loss with their conversion to categorical predictors and (b) variables that are potentially irrelevant to both future regression- or simulation-based approaches to dwell time prediction.

More broadly, this analysis seems to support the utility of this approach to variable selection for simulation design. The goal of isolating particular subsets of variables, whether in the context of deciding the scope of an analytics project or to isolate a particular subset of observations from which train dwell times can reasonably be simulated, is to optimize for the smallest amount of variance with the broadest possible applicability. By isolating variables for which the greatest amount of variance is explained with the fewest predictors, it logically follows that the variables identified

## References

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